Unit

30 Isomerism

Pre-study

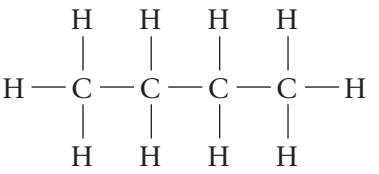
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

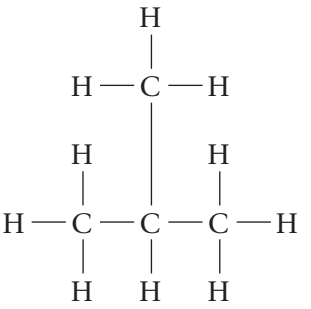
- Structural isomerism — chain isomerism, position isomerism and functional group isomerism
- Stereoisomerism — *cis-trans* isomerism and enantiomerism

30.1 Isomerism

From the study of alkanes, we saw that a compound of the molecular formula C_4H_{10} could have two possible structural formulae:



butane



methylpropane

Compounds which possess the same molecular formula but with their atoms arranged in different ways are called **isomers**. The phenomenon is called **isomerism**.

✓ Isomers are different compounds that have the same molecular formula.

There are two types of isomerism:

- **structural isomerism** — in which atoms are linked in different orders; and
- **stereoisomerism** — in which atoms are linked in the same way but have different spatial arrangements.

isomer 同分異構體　　isomerism 同分異構　　structural isomerism 結構異構　　stereoisomerism 立體異構

Animation
Isomerism

30.2 Structural isomerism

Structural isomerism can be divided into three different types: chain, position and functional group isomerisms (Fig. 30.1).

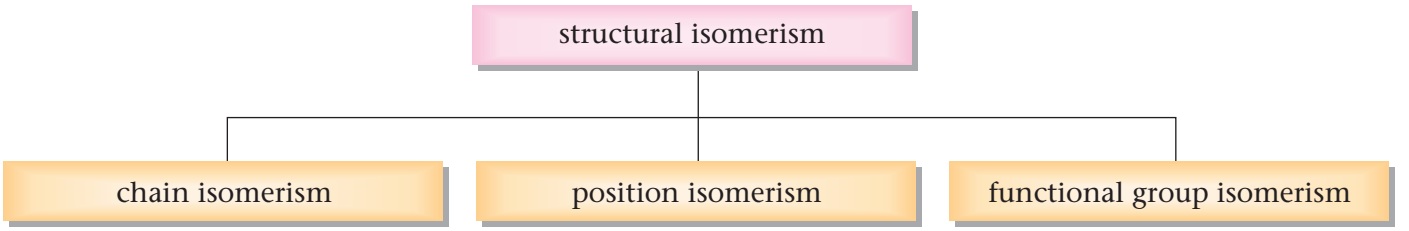


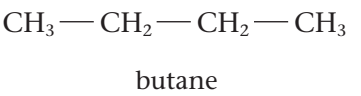
Fig. 30.1 Various ways in which structural isomerism can arise

30.1
Building molecular models of structural isomers.

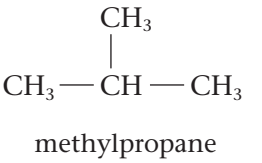
Chain isomerism

✓ **Chain isomers** are isomers that have the same functional groups but different carbon skeletons.

These isomers arise because of the possibility of branching in carbon chains. Butane and methylpropane are chain isomers of C_4H_{10}. In one of them, the carbon atoms lie in a straight chain. In the other, the chain is branched. N1



butane

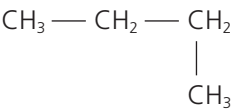


methylpropane

Both isomers belong to the same homologous series, the alkanes. This means they have similar chemical properties.

Isomers often involve branched N2 chains. Many people make the mistake of drawing 90° bonds in a carbon chain rather than branches, thinking that this forms a different isomer.

For example,



is not a third isomer of C_4H_{10} because it can form from butane simply by rotating about a C–C single bond.

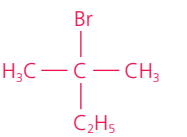
Reference Website

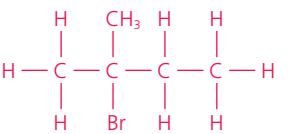
Refer to the following website for the different isomers of pentane:
http://resources.hkedcity.net/resource_detail.php?rid=435840695
(accessed July 2014)

chain isomer 鏈異構體

N1 Refer to page T25.

N2 The following two molecules are identical:





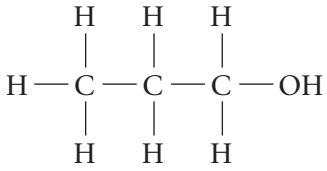
Position isomerism

✓ **Position isomers** are isomers that have the same carbon skeleton and functional group. They differ only in the position of the functional group.

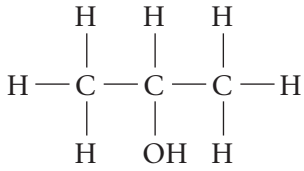
Position isomers are chemically similar because they possess the same functional group.

Position isomers of C_3H_7OH

Two position isomers of the molecular formula C_3H_7OH are shown below.



propan-1-ol

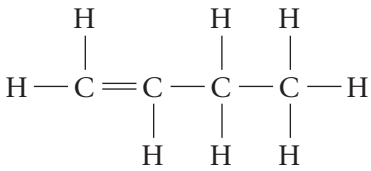


propan-2-ol

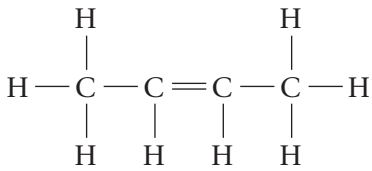
The –OH functional group is situated at two different positions on the carbon chains.

Position isomers of C_4H_8

Two position isomers of the molecular formula C_4H_8 are shown below.



but-1-ene



but-2-ene◂

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The C=C functional group is situated at two different positions on the carbon chains.

◂ The but-2-ene shown here is *cis*-but-2-ene. It has an isomer called *trans*-but-2-ene. We will discuss *cis-trans* isomers later in this unit.

position isomer 位置異構體

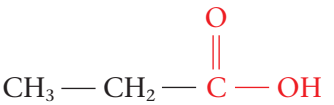
Functional group isomerism

✓ **Functional group isomers** are isomers that have the same molecular formula but contain different functional groups.

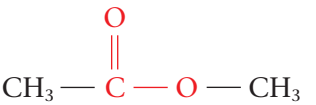
Such isomers have different functional groups and they belong to different homologous series.

Molecular formula $C_3H_6O_2$

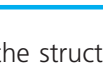
Two functional group isomers of the molecular formula $C_3H_6O_2$ are shown below.



propanoic acid



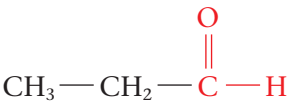
methyl ethanoate

The isomer on the left is a carboxylic acid. Its functional group is –COOH. The isomer on the right is an ester. Its functional group is  . They have very different chemical properties as a result of their different functional groups.

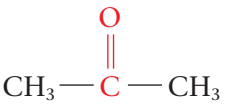
We have discussed the structures and names of carboxylic acids and esters in Unit 29.

Molecular formula C_3H_6O

Two functional group isomers of the molecular formula C_3H_6O are shown below.



propanal



propanone

The isomer on the left is an aldehyde. The isomer on the right is a ketone.

We have discussed the structures and names of aldehydes and ketones in Unit 29.

functional group isomer 官能基異構體

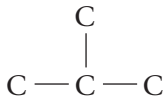
Example 30.1

Q Draw and name the structural isomers of C_4H_9Cl.

A There are two ways of arranging the four carbon atoms:

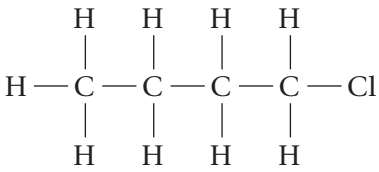


skeleton A

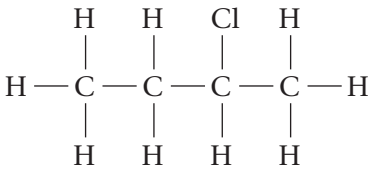


skeleton B

There are two possible positions for the chlorine atom on skeleton A:

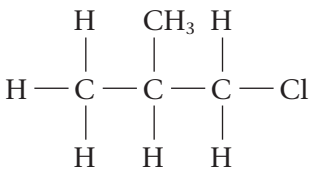


1-chlorobutane

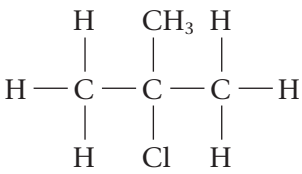


2-chlorobutane

There are also two possible positions for the chlorine atom on skeleton B:



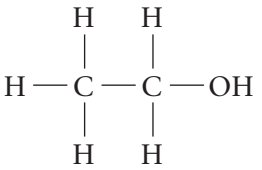
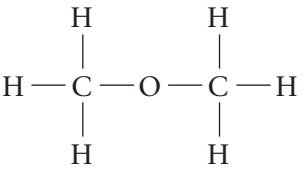
1-chloro-2-methylpropane

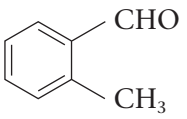
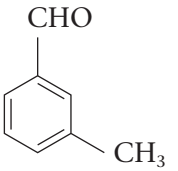


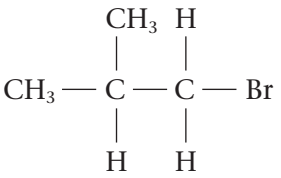
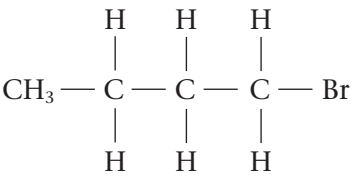
2-chloro-2-methylpropane

Example 30.2

Q For each pair of molecules shown below, classify their relationship as 'chain isomers', 'position isomers' or 'functional group isomers'.

a)  and 

b)  and 

c)  and 

A a) Functional group isomers

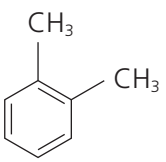
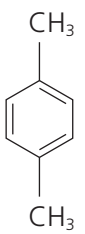
b) Position isomers

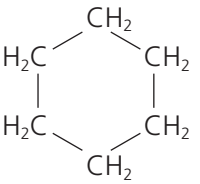
c) Chain isomers

Practice 30.1

1 Draw the structures of the three structural isomers of C_5H_{12}. Give the systematic name of each isomer.
Refer to page T26.

2 For each pair of molecules shown below, classify the relationship as 'chain isomers', 'position isomers' or 'functional group isomers'.

a)  and 
position isomers

b)  and $CH_3CH_2CH{=}CHCH_2CH_3$
functional group isomers

c) $CH_3CH_2CH_2CH_2OH$ and $CH_3CH(CH_3)CH_2OH$
chain isomers

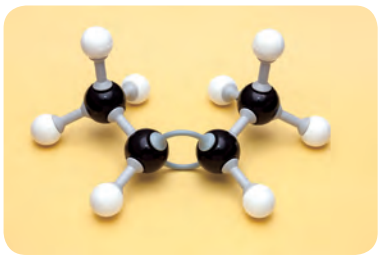
30.3 *Cis-trans* isomerism

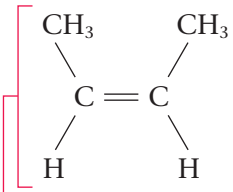
30.2
Building molecular models of *cis-trans* isomers.

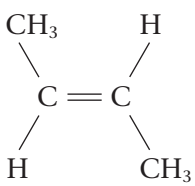
✓ Stereoisomers have their atoms linked in the same way, but they differ in the spatial arrangement of their atoms.

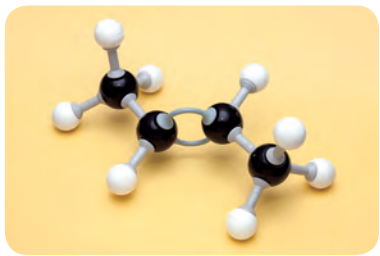
N3 Refer to page T26.

Look at the two forms of but-2-ene shown below. Are they the same? N3









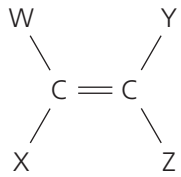
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To turn the second form into the first form, you would have to spin one end of the molecule round in relation to the other end. However, this rotation is restricted by the carbon-carbon double bond. Thus, the two forms cannot be interconverted.

The terms *cis* and *trans* are from Latin in which *cis* means 'on the same side' and *trans* means 'on the other side' or 'across'.

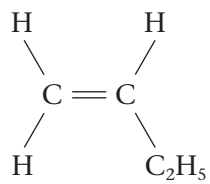
✓ Stereoisomers that have a different arrangement of their atoms in space due to the restricted rotation about a carbon-carbon double bond are ***cis-trans* isomers**.

The following carbon compound exhibits *cis-trans* isomerism if W ≠ X and Y ≠ Z.



Cis-trans isomerism is common in compounds containing a carbon-carbon double bond. Compounds with two different atoms or groups on each carbon atom of the double bond have two alternative structures, which are *cis* and *trans* isomers. N4

Consider the compound but-1-ene:



One of the carbon atoms of the double bond of but-1-ene has two identical hydrogen atoms attached to it. Hence but-1-ene does not exhibit *cis-trans* isomerism.

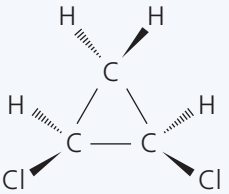
N4 But-2-ene can exhibit *cis-trans* isomerism because
- rotation about the C=C bond is restricted;
- there are different atoms or groups of atoms on each carbon atom of the C=C bond.

cis-trans isomers 順-反異構體

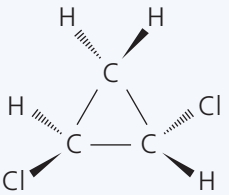
Do you know ?

Cis-trans isomerism in cyclic compounds

Cis-trans isomerism can also occur in cyclic compounds, in which rotation is not possible. For example, 1,2-dichlorocyclopropane can exhibit *cis-trans* isomerism.



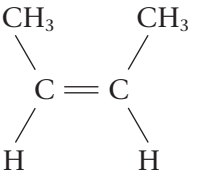
cis isomer



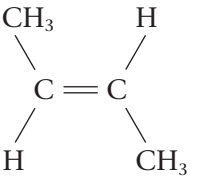
trans isomer

Naming *cis* and *trans* isomers

Take the two different but-2-enes as an example. The form of but-2-ene with the methyl groups on the same side of the double bond is called *cis*-but-2-ene. The form with the methyl groups on opposite sides of the double bond is called *trans*-but-2-ene.



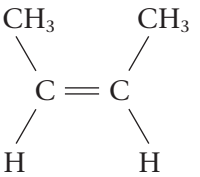
cis-but-2-ene



trans-but-2-ene

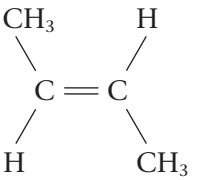
Physical properties of *cis* and *trans* isomers

Cis and *trans* isomers normally have similar chemical properties. However, their physical properties are often quite different. For example, *cis*-but-2-ene and *trans*-but-2-ene have different melting points.



cis-but-2-ene

melting point –139 °C



trans-but-2-ene

melting point –105 °C

The melting point of a carbon compound depends on the strength of intermolecular attractions as well as molecular symmetry. N5

N5 Refer to page T28.

Van der Waals' forces exist in the two forms of but-2-ene and they are of comparable strength.

The *trans* isomer has a more regular and symmetrical structure than the *cis* isomer. Molecules of the *trans* isomer can pack more compactly in the solid. More heat is required to overcome the van der Waals' forces between the molecules. Hence the *trans* isomer has a higher melting point than the *cis* isomer. N6

ASL 2009 Paper 2 Q4(b)(ii)
AL 2009 Paper 2 Q4(a)(ii)(II)

N6 The melting point of *trans*-1,2-dichloroethene is higher than that of *cis*-1,2-dichloroethene.



–50	melting point (°C)	–80

Practice 30.2 Refer to page T29.

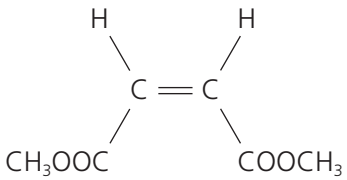
1 Decide whether each of the following compounds exhibit *cis-trans* isomerism. Explain your answer in each case.

a) 1,2-dibromoethene

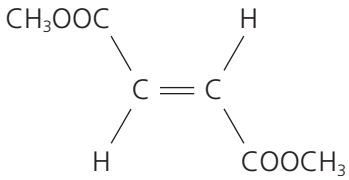
b) hex-1-ene

ASL 2010 Paper 1 Q6(a)(ii)
AL 2010 Paper 1 Q4(a)(ii)

2 The structures and melting points of two isomers are given below:



melting point –19 °C



melting point 102 °C

Explain why the melting point of the first isomer is lower than that of the second isomer.

Being superposable is different from being **superimposable**. Any two objects can be superimposed simply by putting one object on top of the other, whether or not all parts coincide.

30.4 Chirality

When you view your left hand in a mirror, the mirror image of your left hand is a right hand (Fig. 30.2a). Yet your left and right hands cannot be placed on each other so that all parts coincide. In other words, they are not **superposable** (Fig. 30.2b). This happens because a hand is not a symmetrical object. This property is known as **chirality**.

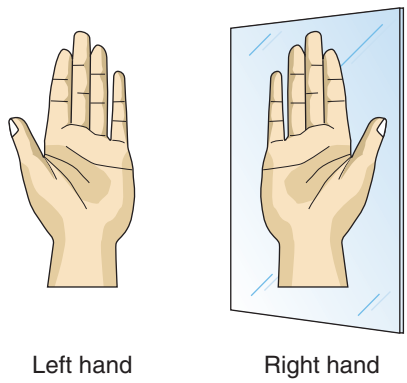


Fig. 30.2a The mirror image of a left hand is a right hand

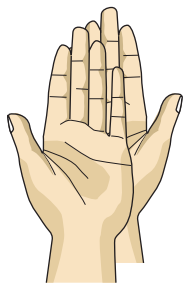


Fig. 30.2b Left and right hands are not superposable

superposable 可疊合的　　superimposable 可疊置的　　chirality 手性

The word chiral comes from the Greek word *cheir*, meaning hand.

A **chiral** object is one that is not superposable on its mirror image. An object that is superposable on its mirror image is **achiral** (meaning not chiral). For example, socks are achiral (Fig. 30.3).



Fig. 30.3 Socks are achiral

30.5 Enantiomers

A molecule can be chiral and so has a non-superposable mirror image. The simplest type of chiral molecules is where four different groups are attached to one carbon atom.

Looking at the chiral molecule of a substituted methane CHFClBr

Consider the simple molecule of a substituted methane CHFClBr. Fig. 30.4a shows a molecular model of CHFClBr (model I). If we hold this model before a mirror, we will see its mirror image (model II).

Animation
Chirality

Reference Website
Refer to the following website for a chirality game:
http://nobelprize.org/nobel_prizes/chemistry/laureates/2001/illpres/game.html
(accessed July 2014)

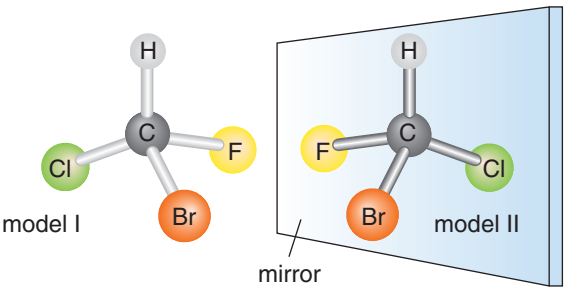


Fig. 30.4a The CHFClBr molecule and its mirror image

Rotation of the mirror image model (i.e. model II) about the C–H bond produces model IIa shown in Fig. 30.4b.

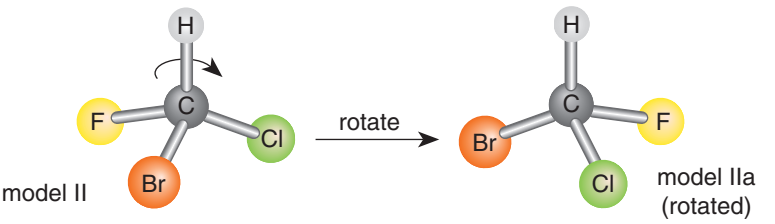


Fig. 30.4b Model II being rotated about the C–H bond

chiral 手性的　　achiral 非手性的

Fig. 30.4c shows the situation when we superimpose models I and IIa (rotated model II). Notice that in the two models, the C, H and F atoms are in the same spatial position but the positions of Cl and Br do not match. The two models and the molecules they represent are not superposable.

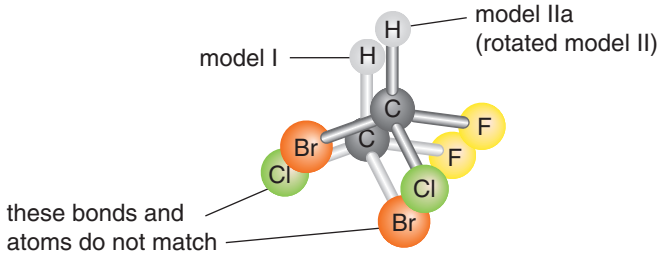


Fig. 30.4c Models I and IIa (rotated model II) are not superposable

The CHFClBr molecule is chiral. The central carbon atom is called the **chiral carbon** (Fig. 30.5). N7 N8

N7 Refer to page T30.

N8 A molecule with a single chiral carbon must be chiral. A molecule with two or more chiral carbons may be chiral or it may not be.

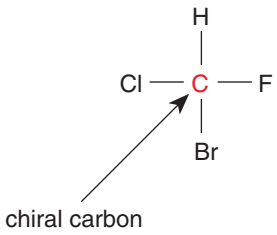


Fig. 30.5 The chiral carbon in the CHFClBr molecule

✓ A chiral molecule and its mirror image are called a pair of **enantiomers**.

✓ Enantiomers are stereoisomers whose molecules are non-superposable mirror images of each other.

chiral carbon 手性碳　　enantiomer 對映異構體

All molecules have mirror images, but they do not all exist as two enantiomers. Consider another substituted methane having two identical hydrogen atoms, CH_2ClBr. Fig. 30.6a shows the model of the molecule and its mirror image.

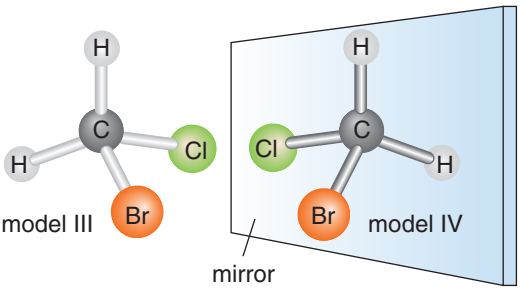


Fig. 30.6a The CH_2ClBr molecule and its mirror image

Fig. 30.6b shows the situation when we superimpose models III and IVa (rotated model IV). The two models and the molecules they represent are identical. The CH_2ClBr molecule has only one form.

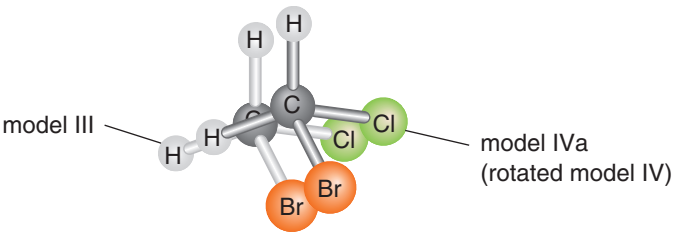
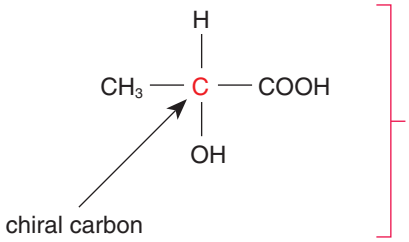


Fig. 30.6b Models III and IVa (rotated model IV) are identical

Looking at the chiral molecule of lactic acid

Lactic acid is produced when milk goes sour. Its chemical formula is $CH_3CH(OH)COOH$. A molecule of lactic acid contains a chiral carbon that has –H, $–CH_3$, –OH and –COOH attached (Fig. 30.7).



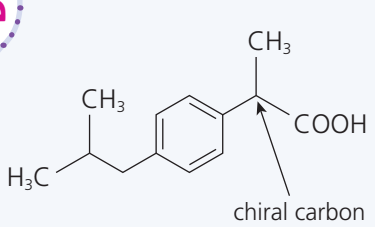
ASL 2011 Paper 1 Q7(a)

Fig. 30.7 The chiral carbon in the molecule of lactic acid

Do you know ?

Ibuprofen

Ibuprofen is a type of anti-inflammatory drug that targets both bone and muscle pain.



structure of ibuprofen

Ibuprofen has one chiral carbon, so two enantiomers are possible. One of the isomers relieves pain much more effectively than the other. Fortunately in the body the less active form of ibuprofen is converted to the pharmacologically active isomer, so the whole dose given to a patient is active. This minimizes any possible side effects. Ibuprofen is sold as a mixture of the two enantiomers.

Game
Identifying isomers

Fig. 30.8 shows the two enantiomers of lactic acid. They are mirror images.

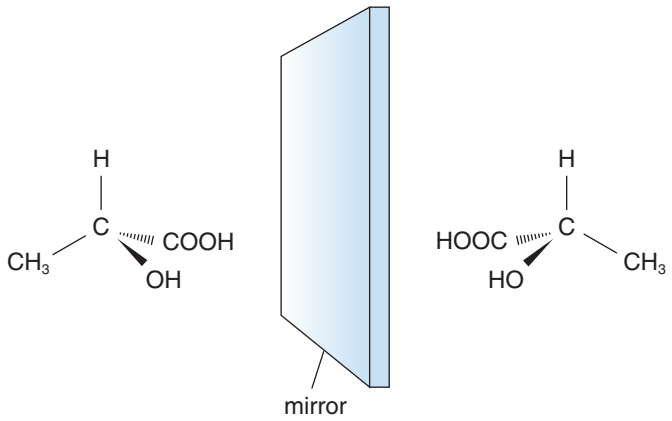


Fig. 30.8 Enantiomers of lactic acid

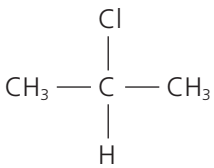
30.6 Test for chirality — plane of symmetry

We have already learnt one aid that helps us in recognizing chiral molecules — the presence of a carbon atom bonded to four different atoms or groups of atoms.

The other aid is that a molecule will not be chiral if it has a **plane of symmetry**, i.e. an imaginary plane that bisects a molecule in such a way that the two halves are mirror reflections of each other.

Molecules having a plane of symmetry in them are symmetrical and achiral. For example, the molecule of 2-chloropropane has a plane of symmetry (Fig. 30.9a) and is achiral. The molecule of 2-chlorobutane does not have a plane of symmetry and is chiral (Fig. 30.9b).

A molecule of 2-chloropropane is achiral. The central carbon atom bears two identical methyl groups.



30.3
Building models of some molecules and determining whether the molecules are chiral.

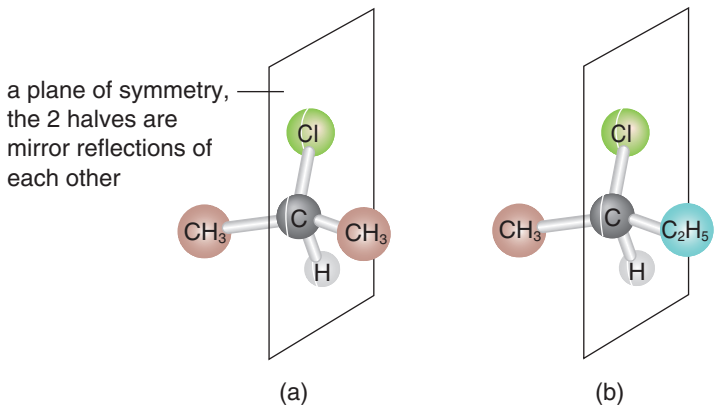


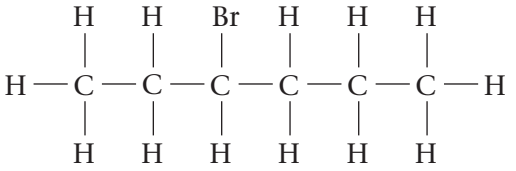
Fig. 30.9 (a) A molecule of 2-chloropropane has a plane of symmetry and is achiral
(b) A molecule of 2-chlorobutane does not have a plane of symmetry and is chiral

plane of symmetry 對稱面

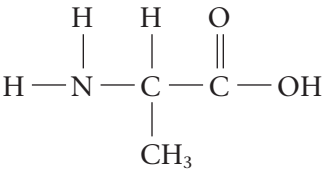
Example 30.3

Q Consider the following chiral molecules:

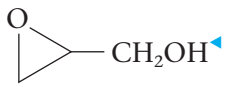
a)



b)



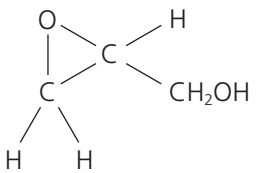
c)



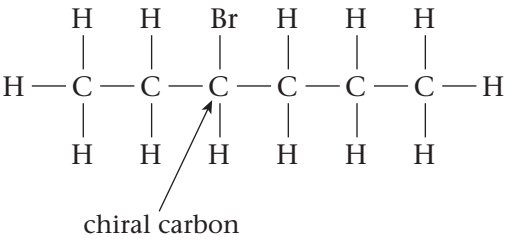
ASL 2007 Paper 1 Q8(a)
ASL 2009 Paper 1 Q3(c)
AL 2005 Paper 2 Q5(b)(i)
AL 2007 Paper 1 Q6(a)

Mark the chiral carbon in each molecule.

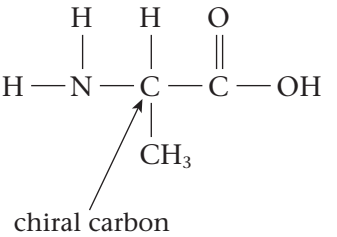
It is easier to identify the chiral carbon if we draw the compound in the following way:



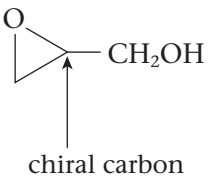
A a)



b)



c)



Practice 30.3

1 Which of the following structural formulae can represent two enantiomers? Explain your choice.

$CH_3CH_2CH_2CH(OH)CH_3$

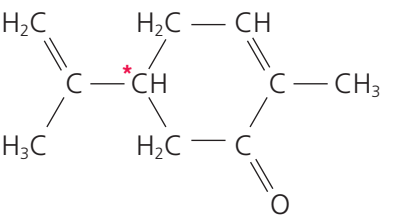
$CH_3CH_2CHClCH_2CH_3$

$CH_3CH_2CH_2C(OH)_2CH_2CH_2Cl$

$CH_3CH_2CH_2CH(OH)CH_3$ can represent two enantiomers. It has a chiral carbon atom.

$CH_3CH_2CHClCH_2CH_3$ and $CH_3CH_2CH_2C(OH)_2CH_2CH_2Cl$ have no enantiomer. They have no chiral carbon.

2 Carvone is a chiral molecule found in the oil of caraway seeds. Its structure is shown below:



Mark the chiral carbon with an asterisk (*).

30.7 Distinguishing the enantiomers of a chiral compound

N9 Refer to page T30.

Identifying and distinguishing enantiomers are difficult as they have the same physical and chemical properties (except in chemical reactions with chiral reagents). N9

Fortunately, scientists discovered that enantiomers of a chiral compound have different effects on **plane-polarized light**. These substances are said to be **optically active**.

Plane-polarized light

Light is a form of electromagnetic radiation. A beam of light consists of two mutually perpendicular oscillating fields — an oscillating electric field and an oscillating magnetic field (Fig. 30.10).

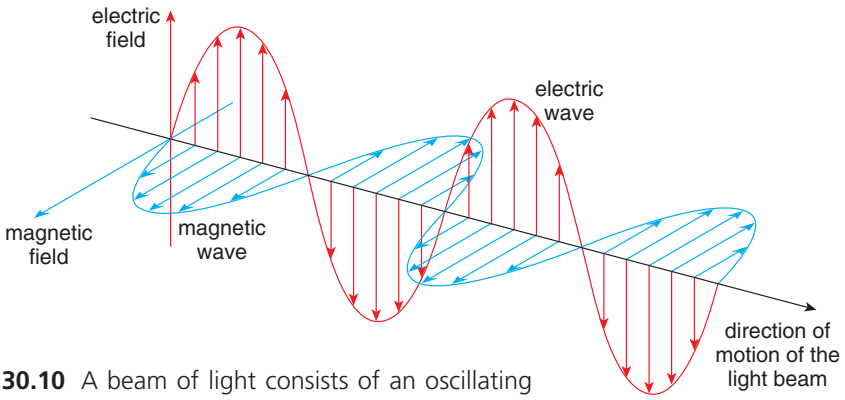


Fig. 30.10 A beam of light consists of an oscillating electric field and an oscillating magnetic field

In a beam of ordinary light, oscillation of the electric field occurs in all possible planes perpendicular to the *direction of propagation* (Fig. 30.11). (The same would be true of the magnetic field.)

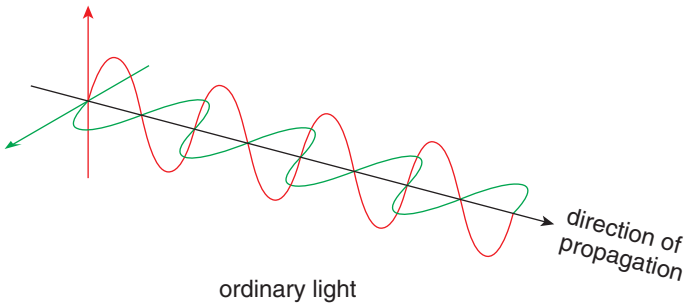


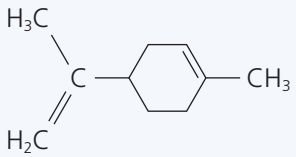
Fig. 30.11 Oscillation of the electric field of ordinary light occurs in all possible planes perpendicular to the direction of propagation

Do you know ?

Smells of limonene

Although enantiomers have identical chemical properties, they do interact differently with the enantiomers of other chiral molecules. Hence some enantiomers do not smell and taste the same, because they interact differently with the chiral taste and smell receptors.

For example, enantiomers of limonene produce different smells. One enantiomer gives the smell of oranges and the other gives the smell of lemons.



structure of limonene

Reference Website

Refer to the following video clip on how enantiomers of a chiral compound perturb plane-polarized light:
http://www.youtube.com/watch?v=uD9j3nbaHsE&feature=related
(accessed July 2014)

plane-polarized light 平面偏振光 optically active 旋光性 direction of propagation 傳播方向

When ordinary light is passed through a polarizer, the polarizer interacts with the electric field. The electric field of the light that emerges from the polarizer (and the magnetic field perpendicular to it) is oscillating only in one plane, for example, the vertical plane (Fig. 30.12). Such light is called plane-polarized light.

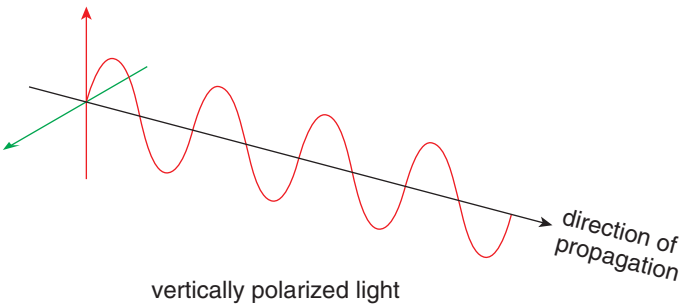


Fig. 30.12 The plane of oscillation of the electric field of a beam of plane-polarized light

Polarimeter

The angle through which the plane of polarization is rotated depends on:

- the nature of the enantiomer;
- the length of the sample tube; and
- the concentration of the enantiomer in the solution.

To study the interaction between plane-polarized light and an optically active compound, we use a polarimeter shown in Fig. 30.13. A beam of unpolarized light first passes through a polarizer, and then through a sample tube containing a solution of the compound. As the polarized light passes through the sample tube, its **plane of polarization** is rotated either clockwise or anticlockwise.

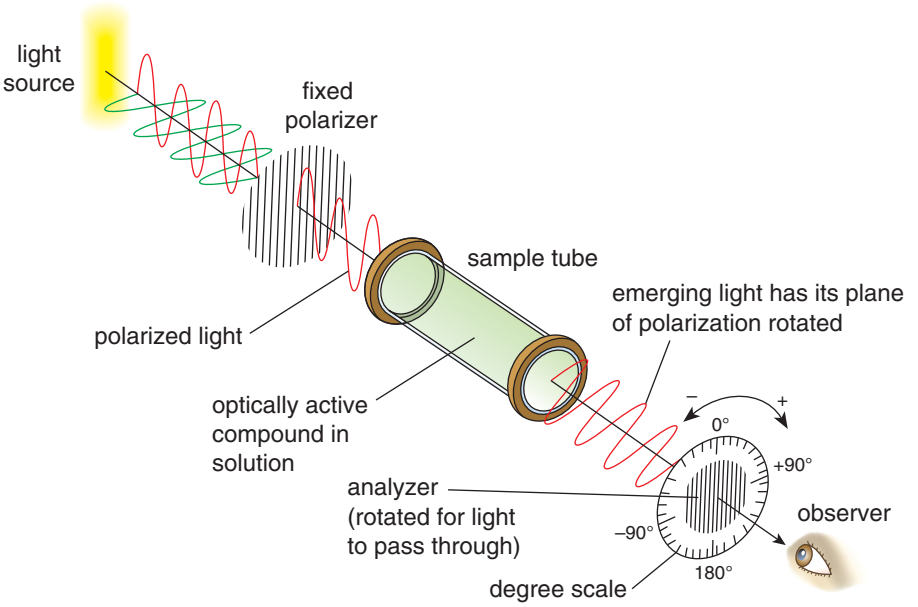


Fig. 30.13 The principal working part of a polarimeter

plane of polarization 偏振面

Enantiomers of a chiral compound always rotate the plane of polarization of a beam of plane-polarized light by the same amount, but in opposite directions, clockwise in one case (Fig. 30.14a), and anticlockwise in the other (Fig. 30.14b).

AL 2009 Paper 2 Q5(b)(i)(III)

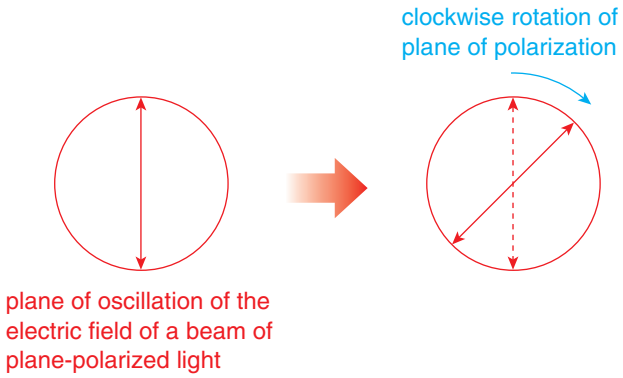


Fig. 30.14a Clockwise rotation of plane of polarization

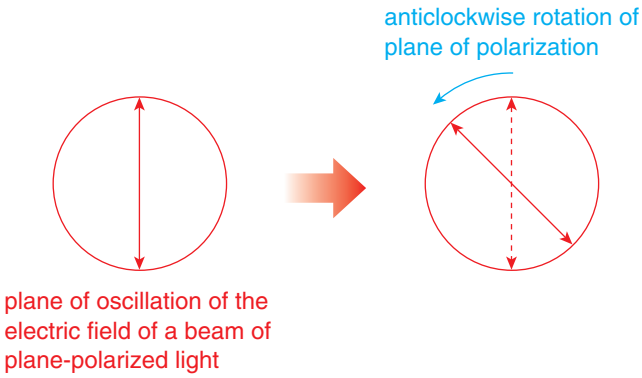


Fig. 30.14b Anticlockwise rotation of plane of polarization

Example 30.4

Q Compound X has the following structural formula:

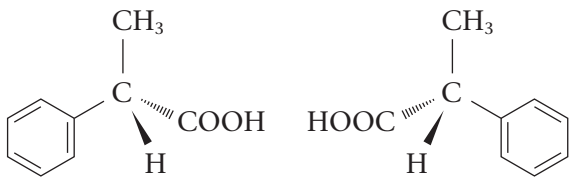


The above structural formula can represent two stereoisomers. N10 N11

a) Draw the three-dimensional structures of the two stereoisomers.

b) State a physical property which is different for the two stereoisomers. N12

A a)



b) Both isomers show optical rotation. One of them rotates the plane of polarization of a beam of plane-polarized light clockwise while the other anticlockwise.

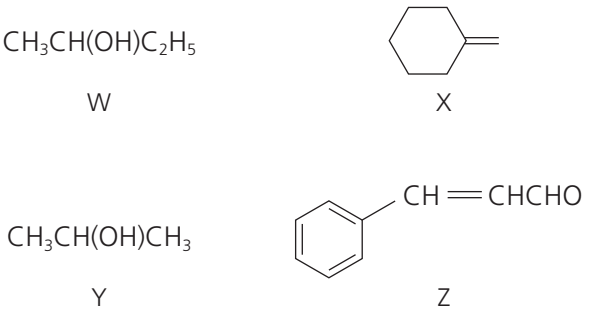
N10 Refer to page T31.

N11 Refer to page T31.

N12 Examination questions often ask students to suggest a different physical property between a pair of enantiomers.

Practice 30.4

Consider the structural formulae W, X, Y and Z shown below:



a) i) Which structural formula represents a compound that exhibits *cis-trans* isomerism? Z

ii) Draw the *trans*-isomer of the compound.
Refer to the next page.

b) i) Which structural formula represents two compounds that are non-superposable mirror images? W

ii) Draw a three-dimensional structure for each of the two compounds.
Refer to the next page.

iii) Decide whether the two compounds can be separated by fractional distillation. Explain.
No. They have the same boiling point.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

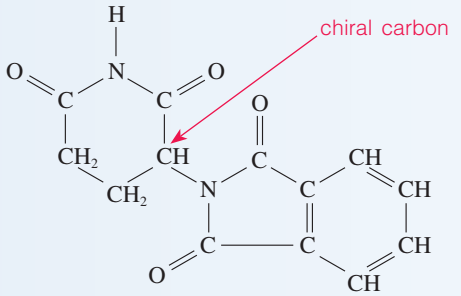
Chemistry Magazine

The thalidomide tragedy

In 1954, a German pharmaceutical company developed a new drug called thalidomide. In the late 1950s and the early 1960s, doctors prescribed thalidomide to prevent morning sickness in pregnant women.

In this period, 12 000 babies worldwide were born with unusual birth defects — their limbs did not grow properly. Eventually it was realized that thalidomide was the cause of the problem (Fig. 30.15).

Thalidomide is a chiral compound and the drug is a mixture of its two enantiomers. While one of the enantiomers has the required therapeutic effect, unfortunately its mirror image affects the limb development of a foetus. The drug was withdrawn in 1961.



structure of thalidomide

Even after the ban in 1961, pharmacologists continued to work with thalidomide. It is found that the drug can be used to relieve other symptoms. It appears that so long as it is not given in pregnancy, it is a safe and potentially useful drug.

Questions

1 Identify the chiral carbon in a molecule of thalidomide.

2 Discuss the advantages of producing a drug that contains a single enantiomer for patients and for drug companies. Refer to page T32.

3 Suggest ONE way of drug testing today which ensures that the thalidomide tragedy is unlikely to happen again. Refer to page T32.



Fig. 30.15 Defects caused by thalidomide

Have you mastered?

Key terms

isomer 同分異構體 47 isomerism 同分異構 47 structural isomerism 結構異構 47

stereoisomerism 立體異構 47 chain isomer 鏈異構體 48 position isomer 位置異構體 49

functional group isomer 官能基異構體 50 *cis-trans* isomerism 順-反異構 53

cis-isomer 順式異構體 53 *trans*-isomer 反式異構體 53 superposable 疊合 55 chirality 手性 55

chiral 手性的 56 achiral 非手性的 56 chiral carbon 手性碳 57 enantiomer 對映異構體 57

plane of symmetry 對稱面 59 plane-polarized light 平面偏振光 61 optically active 旋光性 61

plane of polarization 偏振面 62

Checklist

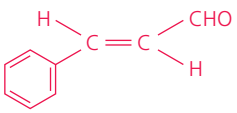
After studying this unit, you should be able to

- ☐ recognize and predict the existence of structural isomerism, including chain isomerism, position isomerism and functional group isomerism;
- ☐ recognize the existence of *cis-trans* isomerism in acyclic carbon compounds resulting from restricted rotation about a C=C bond;
- ☐ predict the structures of the structural or *cis-trans* isomers of given carbon compounds;
- ☐ recognize the existence of enantiomerism in compounds with only one chiral carbon.

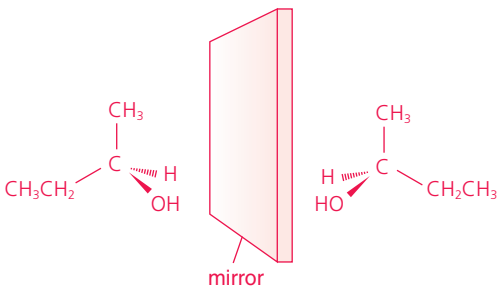
(Put a '✔' in the box if you have acquired the knowledge concerned.)

Practice 30.4

a) ii)



b) ii)



Summary

1 The following charts show the classification of isomers.

isomers
different compounds that have the same molecular formula

structural isomers
atoms are linked in different orders

stereoisomers
atoms are linked in the same way but with different spatial arrangements

structural isomers

chain isomers
isomers with the same functional group but different carbon skeletons
e.g.

$CH_3CH_2CH_2CH_2CH_3$

and

$CH_3CHCH_2CH_3$ (with CH_3 branch)

position isomers
isomers with the same carbon skeleton and functional group, but the position of the functional group is different
e.g.

$CH_3CH_2CH_2CH_2OH$

and

$CH_3CHCH_2CH_3$ (with OH)

functional group isomers
isomers with the same molecular formula but different functional groups
e.g.

$CH_3CH_2C(=O)-OH$

and

$CH_3C(=O)-O-CH_3$

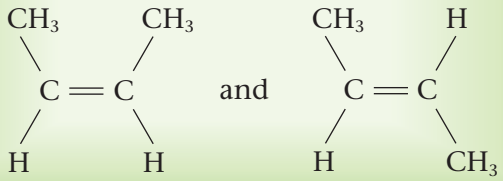
Have you mastered?

stereoisomers

cis-trans isomers

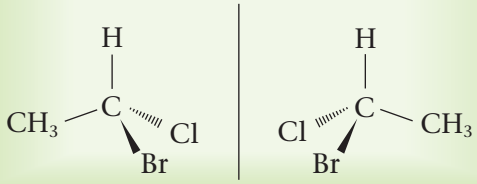
stereoisomers that have a different arrangement of their atoms in space due to restricted rotation about a carbon-carbon double bond

e.g.



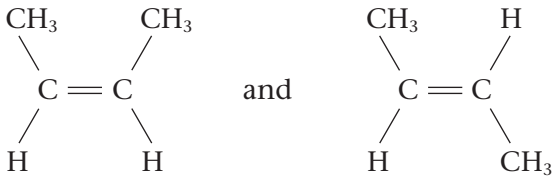
enantiomers

stereoisomers whose molecules are non-superposable mirror images of each other



2 Compounds with two different groups on each carbon atom of the C=C bond have two alternative structures, which are *cis-trans* isomers.

e.g.



cis-but-2-ene *trans*-but-2-ene

3 *Cis-trans* isomers normally have similar chemical properties. However, their physical properties are often quite different.

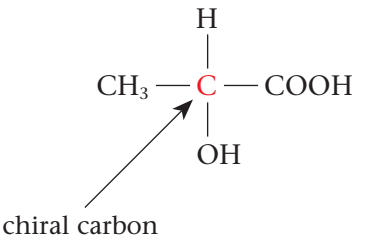
The melting point of *trans*-but-2-ene is higher than that of *cis*-but-2-ene as the *trans* isomer has a more regular and symmetrical structure, hence more heat is required to overcome the van der Waals' forces between the molecules.

4 A chiral molecule and its mirror image are called a pair of enantiomers.

Have you mastered?

5 The simplest type of chiral molecules is where four different groups are attached to one carbon atom. Such a carbon atom is called a chiral carbon.

e.g.



6 A pair of enantiomers have the same physical and chemical properties, except

a) their reactions with chiral reagents; and

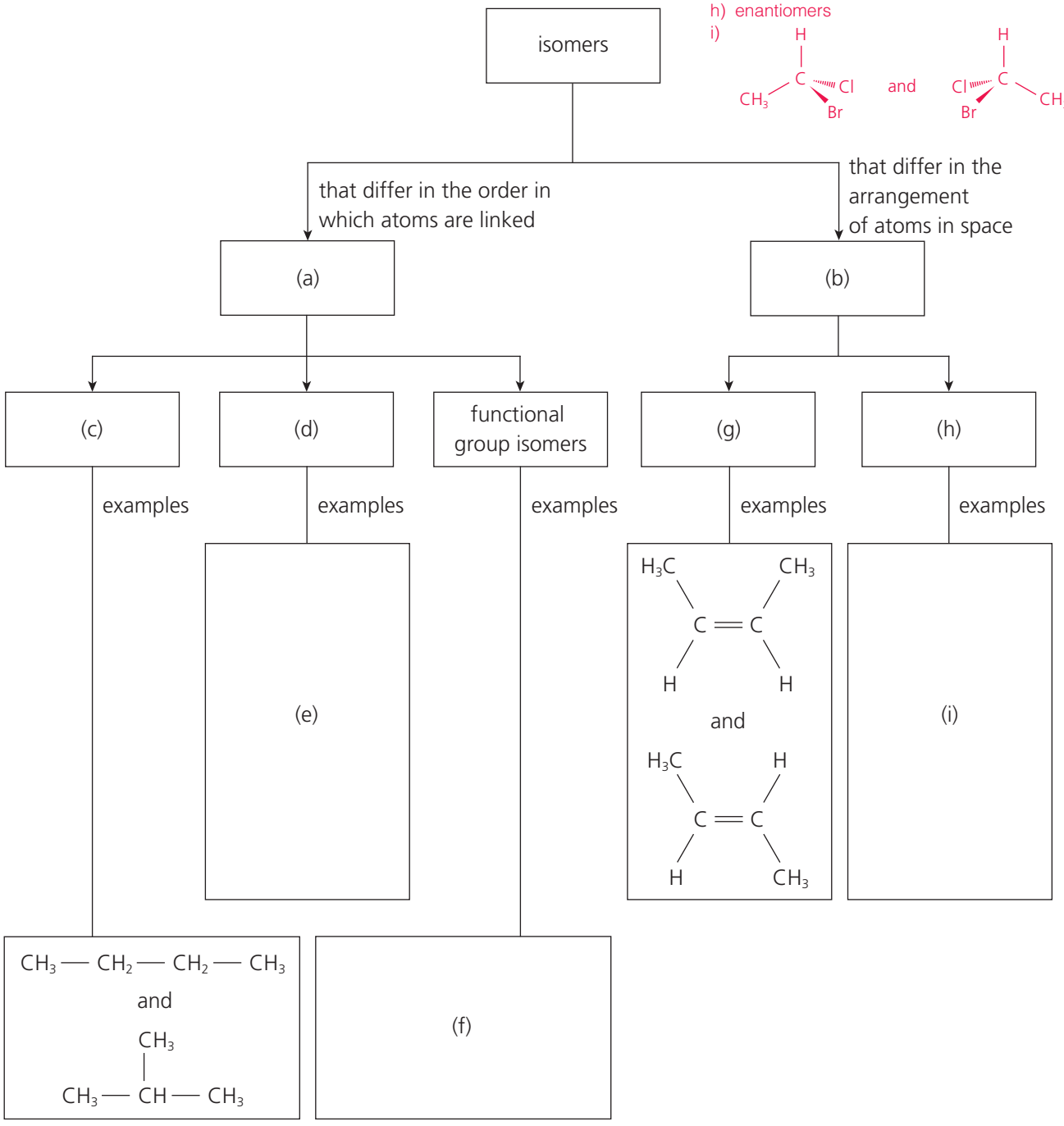
b) they rotate the plane of polarization of plane-polarized light in opposite directions.

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

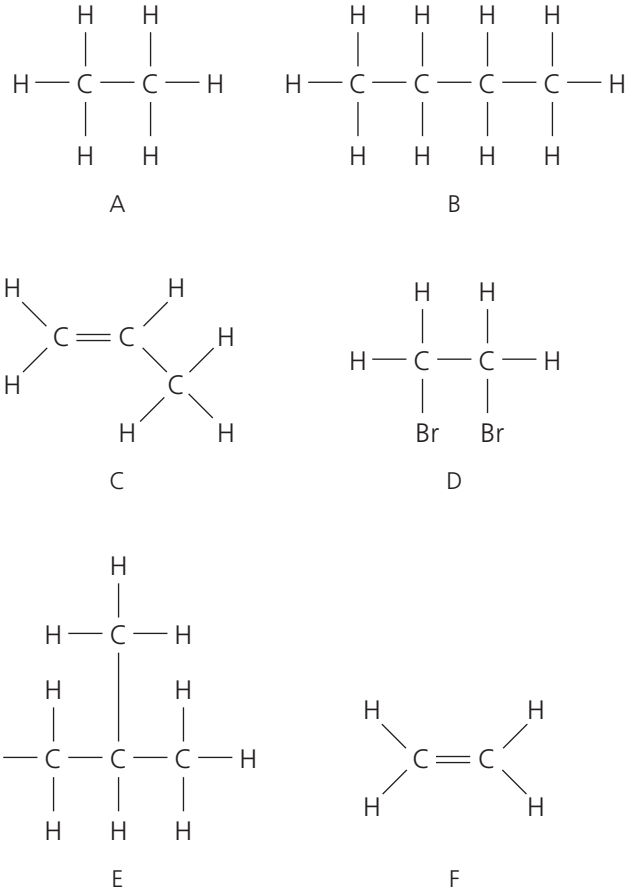
Part I Knowledge and understanding

1 Complete the following concept map.



a) structural isomers
b) stereoisomers
c) chain isomers
d) position isomers
e) $CH_3CH_2CH_2CH_2OH$ and $CH_3CHCH_2CH_3$ (OH)
f) $CH_3CH_2C(=O)OH$ and $CH_3C(=O)OCH_3$
g) *cis-trans* isomers
h) enantiomers
i)

2 The structures of some carbon compounds are shown below.



a) Give the letters of two compounds that are isomers of each other. B, E

b) Give the letters of two compounds that are members of the same homologous series but have different molecular formulae.

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2H, Nov. 2009, 5(c)–(d))

b) Any one of the following:
- A, B, E (Any two)
- C, F

Part II Multiple choice questions

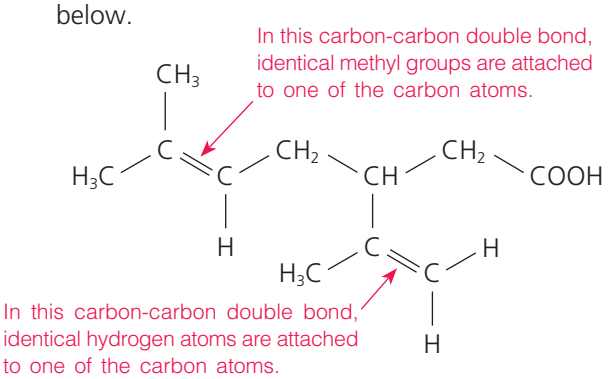
5 How many structural isomers does the alkane C_5H_{12} have?

Refer to page T33 for explanation.

A 4
(B) 3
C 2
D 1

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 11)

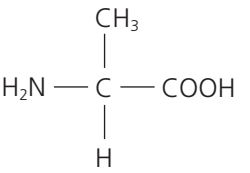
3 The structural formula of a compound is shown below.



a) Name TWO different functional groups in the compound. Carbon-carbon double bond
Carboxyl group

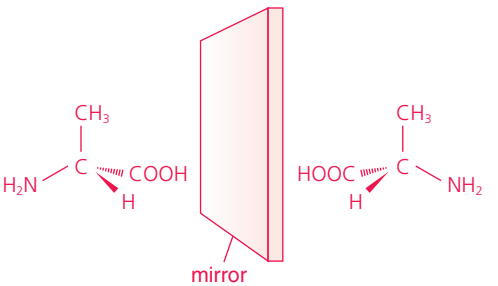
b) Why does the compound NOT exhibit *cis-trans* isomerism?

4 The structural formula shown below can represent two enantiomers.



a) What structural feature causes the compound to have enantiomers? The molecule contains one carbon atom bonded to four different atoms or groups of atoms.

b) Draw the three-dimensional structures of the two enantiomers to show how they are related.



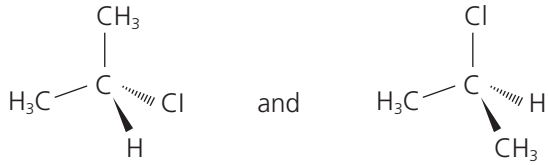
6 What is the number of structural isomers that are carboxylic acids with the molecular formula $C_4H_8O_2$?

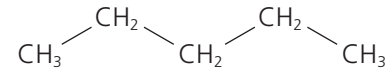
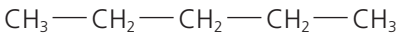
(A) 2
B 3
C 4
D 5

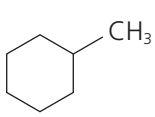
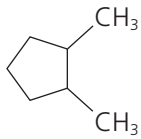
The structural isomers that are carboxylic acids with the formula $C_4H_8O_2$ are shown below:

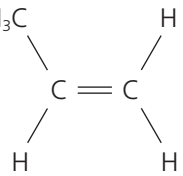
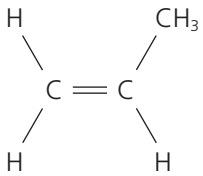


7 Which one is a pair of structural isomers?

A  and

B  and 

(C)  and 

D  and 

8 Which one is a pair of functional group isomers?
Refer to page T33 for explanation.

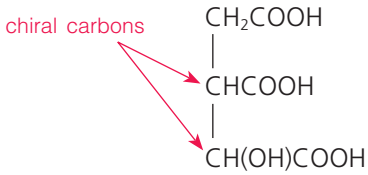
A $CH_3COOCH_2CH_3$ and $CH_3CH_2COOCH_3$
B $(CH_3)_2CHCH(CH_3)_2$ and $(CH_3)_3CCH_2CH_3$
(C) $H_2NCH_2CH_2CH_2COOH$ and $H_2NCH_2COOCH_2CH_3$
D $ClCH_2CH_2CH{=}CH_2$ and $CH_3CH{=}CHCH_2Cl$

9 Which of the following alkenes exhibits *cis-trans* isomerism?

A $H_3CCH{=}C(CH_3)_2$
B $(CH_3)_2C{=}CH_2$
C $H_2C{=}CHCH_2CH_3$
(D) $H_3CCH{=}CHCH_3$

(Edexcel Advanced GCE, Unit 1, Jun. 2009, 8)

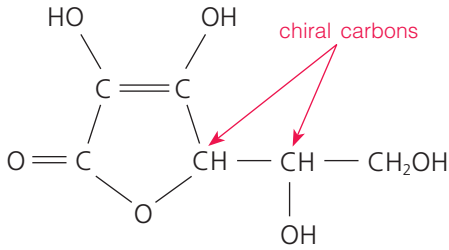
10 The diagram below shows the structure of a carbon compound:



How many chiral carbons are there in one molecule of the compound?

A 1
(B) 2
C 3
D 4

11 The diagram below shows the structure of vitamin C.



How many chiral carbons are there in one molecule of vitamin C?

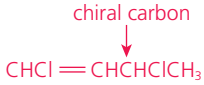
(A) 2
B 3
C 4
D 5

12 Which of the following has both enantiomers and *cis-trans* isomers?

A $ClCH_2CHClCH{=}CH_2$
B $CH_2{=}CClCH_2CH_2Cl$
C $ClCH_2CH{=}CHCH_2Cl$
(D) $CHCl{=}CHCHClCH_3$

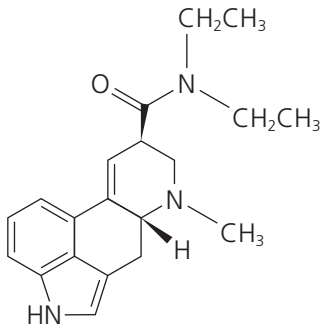
(Edexcel Advanced GCE, Unit 4, Jan. 2011, 9)

$CHCl{=}CHCHClCH_3$ contains one chiral carbon. It has enantiomers.



$CHCl{=}CHCHClCH_3$ has two different atoms or groups of atoms on each carbon atom of the carbon-carbon double bond. It has *cis-trans* isomers.

13 Lysergic acid diethylamide (LSD) is a stimulant drug with the following structure:

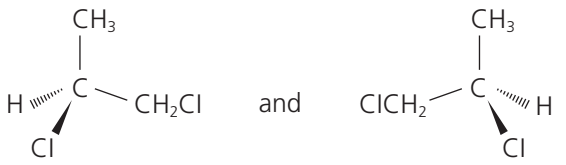


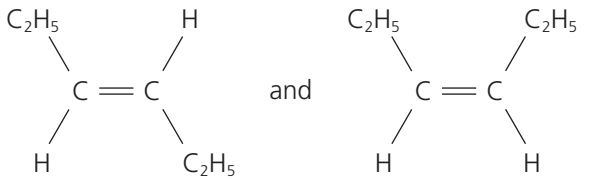
Which one of the following statements about LSD is correct?

A It has one chiral centre and possesses an amine functional group.
B It has one chiral centre and possesses an alkene functional group.
C It has two chiral centres and possesses an amide functional group.
D It has two chiral centres and possesses a ketone functional group.

(HKASLE, Paper 1, 2009, 3(c))

14 In which one of the following pairs do the isomers have identical boiling points?

A $CH_3CH_2CH_2CH_2OH$ and $(CH_3)_2CHCH_2OH$
B $CH_3(CH_2)_4CH_3$ and $(CH_3)_2CHCH(CH_3)_2$
C 

D 

Option C — The two isomers are enantiomers. They have identical boiling point.

15 The structures of two carbon compounds are shown below.



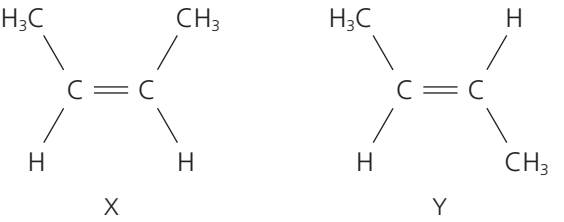
Which of the following statements concerning these two compounds is / are correct?

(1) They have the same relative molecular mass.
(2) They are *cis-trans* isomers.
(3) They are both miscible with water in all proportions.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) Both compounds have the molecular formula $C_3H_6O_2$. Thus, they have the same relative molecular mass.
(2) They are structural isomers.
(3) CH_3CH_2COOH is miscible with water in all proportions but CH_3COOCH_3 is not.

16 The structures of two carbon compounds, X and Y, are shown below:



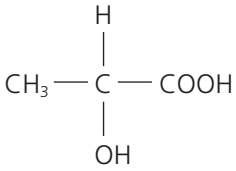
Which of the following statements concerning X and Y is / are correct?

(1) X and Y are stereoisomers.
(2) Both are optically active.
(3) The melting point of X is lower than that of Y.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) X and Y are cis-trans isomers.
(2) Both are optically inactive.
(3) The melting point of X (the cis isomer) is lower than that of Y (the trans isomers).

17 The structure formula shown below can represent two isomers.



Which of the following statements concerning the two isomers is / are correct?

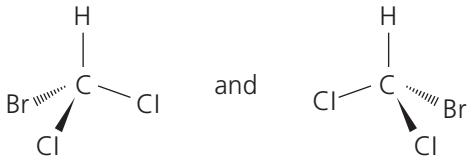
(1) They are *cis-trans* isomers.
(2) They have the same melting points.
(3) They have different effects on plane-polarized light.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

The two isomers are enantiomers.

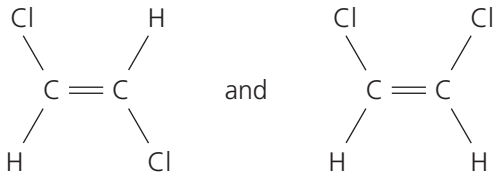
18 Which of the following pairs is / are isomers of each other?

(1)



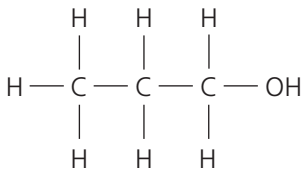
are identical molecules

(2)

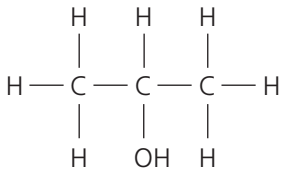


are *cis-trans* isomers

(3)



and



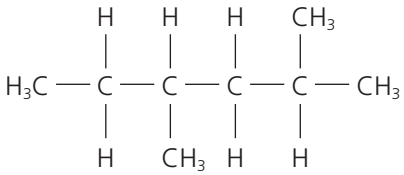
are structural isomers

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

19 a) The structure shown below is one of several structural isomers of octane.



Give the meaning of the term 'structural isomerism'.

Name this isomer and state its empirical formula.

b) Suggest why the branched chain isomer shown above has a lower boiling point than octane.

(AQA Advanced Subsidiary GCE, Unit 1, Jan. 2011, 6(c)–(d))

a) Structural isomers — compounds with the same molecular formula but differs in the order in which atoms are linked
Name of isomer — 2,4-dimethylhexane
Empirical formula — C_4H_9

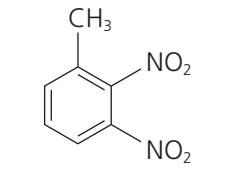
b) The molecule of the branched chain isomer is more compact. The molecular surface area available for coming into contact with neighbouring molecules is smaller.

Hence the van der Waals' forces in the branched chain isomer are less strong.

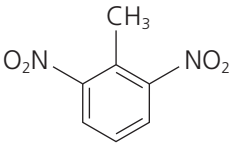
20 Methylbenzene, $C_6H_5CH_3$, is an aromatic hydrocarbon and is used widely as a solvent. It is readily nitrated and it can form mono-, di-, or tri-nitromethylbenzenes.

There are six possible structural isomers of $CH_3C_6H_3(NO_2)_2$ that are dinitromethylbenzenes. Four of the isomers are shown below.

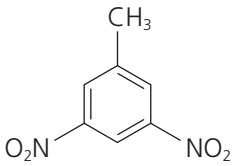
Draw the structures of the other two isomers in the boxes provided.



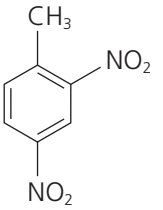
isomer 1



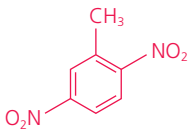
isomer 2



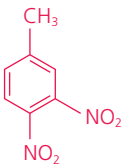
isomer 3



isomer 4



isomer 5



isomer 6

(OCR Advanced GCE, Chem. A, F324, Jan. 2011, 1(b))

21 Pent-1-ene is a member of the alkene homologous series.

a) Pent-1-ene can be separated from other alkenes.

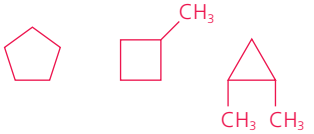
State the physical property of alkenes that allows them to be separated from a mixture by fractional distillation.

a) Different boiling points

b) i) Structural isomers — compounds with the same molecular formula but differs in the order in which atoms are linked

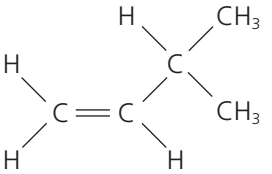
ii) 3-methylbut-1-ene

iii) Any one of the following:



b) i) State the meaning of the term 'structural isomerism'.

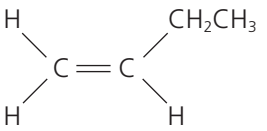
ii) Name the branched chain isomer of pent-1-ene shown below.



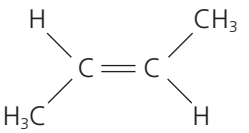
iii) Draw the structure of a functional group isomer of pent-1-ene.

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2010, 6(a)–(b))

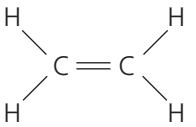
22 Alkenes B, C, D, E and F are shown below.



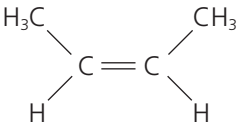
B



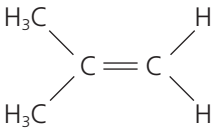
C



D



E



F

You will have to refer to these alkenes throughout the question.

Many alkenes show *cis-trans* isomerism.

a) Explain why *cis-trans* isomerism is shown in some alkenes. Refer to page T33.

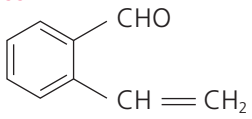
b) Which two alkenes are a pair of *cis-trans* isomers? C, E

Choose from B, C, D, E and F.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2010, 5(b))

23 Consider an aldehyde A with the following structure.

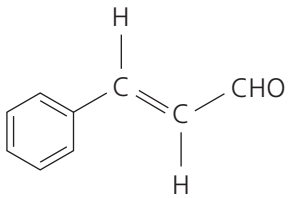
Refer to page T33.



aldehyde A

a) Draw the structures of two position isomers of A.

b) Compound B shown below has the same molecular formula as A. Compound B has a stereoisomer.



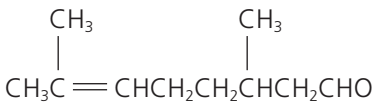
compound B

i) What is meant by the term 'stereoisomer'?

ii) Name the type of stereoisomerism shown by compound B.

iii) Draw the structure of the stereoisomer of compound B.

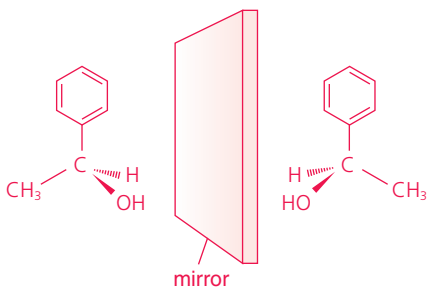
24 A naturally occurring organic compound has the following structure:



On the above structure, circle the chiral carbon centre(s) in this compound.

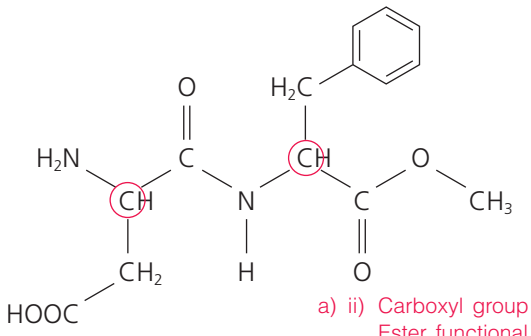
(HKASLE, Paper 2, 2012, 8(a))

26 c)



25 The addition of sucrose, table sugar, to food and drink has been linked to the increased risk of obesity and insulin resistance. Aspartame is used as an alternative to sugar.

The structure of aspartame is shown below.



aspartame

a) ii) Carboxyl group
Ester functional group
Amide functional group
Amine functional group

a) Aspartame contains five functional groups including the benzene ring, and has two chiral carbon atoms.

i) Circle the TWO chiral carbon atoms on the structure above.

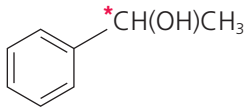
ii) Name the FOUR functional groups, other than the benzene ring, in aspartame.

b) Some artificial sweeteners commonly available many years ago have now been withdrawn from use.

Suggest why. Adverse side effect / toxicity / irritation

(OCR Advanced GCE, Chem. A, F324, Jun. 2011, 6(a), (c))

26 The structural formula shown below can represent two isomers.



key:
* chiral carbon

a) Name the type of isomerism involved.
Enantiomerism

b) Mark on the structural formula the structural feature that is responsible for this type of isomerism.

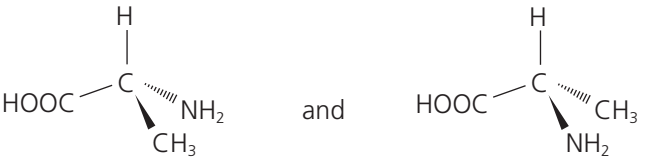
c) Draw three-dimensional structures of the two isomers.

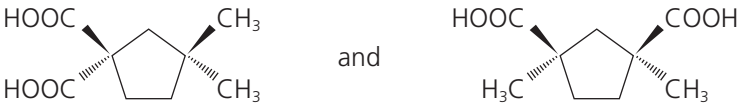
d) State ONE difference in physical properties of the two isomers. They rotate the plane of polarization of plane-polarized light in opposite directions.

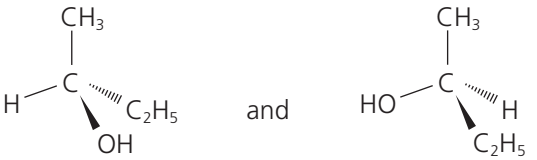
e) Decide whether the two isomers can be separated by fractional distillation. Explain.
No. They have the same boiling point.

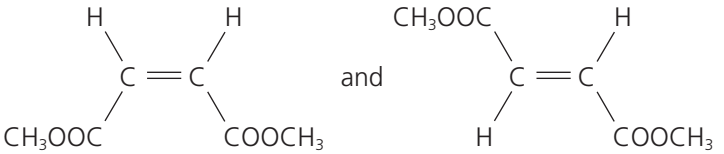
27 For each of the following pairs of molecules, identify their relationship as identical, structurally isomeric, *cis-trans* isomeric or enantiomeric.

a)  identical

b)  enantiomeric

c)  structurally isomeric

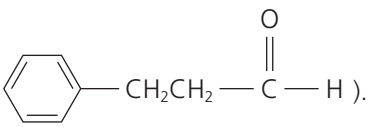
d)  identical

e)  *cis-trans* isomeric

28 Draw the structure of each of the following carbon compounds.

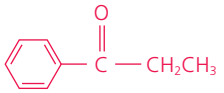
a) A hydrocarbon that is a chain isomer of methylpropene but does not exhibit *cis-trans* isomerism.

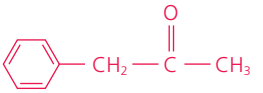
b) A hydrocarbon that is a position isomer of hex-2-ene.

c) A ketone that is a structural isomer of 3-phenylpropanal ().

d) An alcohol that is a structural isomer of butan-1-ol and optically active.

a) $CH_3CH_2CH=CH_2$
b) $CH_3CH_2CH=CHCH_2CH_3$
c) Any one of the following:





d) 